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Practical Assignment 1 - Part-of-speech (PoS) Tagging

Natural Language Understanding.
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Abstract

This brief report explains the process we follow in order to achieve a suitable model for PoS tagging, as well as the results of the test performed for the 3 different languages.

Contents

1 Introduction

Part-of-Speech (POS) tagging is a fundamental step in Natural Language Understanding (NLU), where each word in a sentence is assigned a grammatical category (noun, verb, adjective, etc.). This process supports downstream tasks such as parsing, named entity recognition, and machine translation.
It also present several intrinsic challenges, such as ambiguity, many words vary its POS tag depending on the sentence context (book as noun or verb). Correct tagging often depends on both preceding and following words (context dependency). Unseen words lead to uncertainty in prediction (Data Sparsity). Also we encounter difficulties dealing with morphological richness and grammatical structure across different languages, complicating generalization.

2 Our Approach

To tackle these challenges, we developed a sequence labeling model based on a Bidirectional Long Short-Term Memory (BiLSTM) network with an embedding layer and a TimeDistributed output layer. The model was tested on English, Spanish, and Italian corpora, ensuring multilingual robustness. In order to encode the input for our model we use a "Tokenizer", we also tried with the "Textvectorizer" function but the results were worse.

2.1 Architecture Overview

Our architecture is composed of the following layers:
Embedding Layer: Converts each word index into a dense, low-dimensional vector representation that encodes semantic and syntactic relationships. This layer allows the model to generalize across morphologically similar words, improving multilingual performance.
Bidirectional LSTM Layer: A BiLSTM process input sequences in both forward and backward directions, ensuring that each token representation incorporates context from both preceding and following words. This suits perfectly our task, this type of layer handles ambiguous words more effectively, adapts well to morphologically rich languages, and captures long-range dependencies in both temporal directions.

TimeDistributed Dense Layer: Applies a fully connected layer independently to each timestep while sharing weights across the sequence. This ensures that each contextualized word vector is projected into a probability distribution of the POS tag.
Softmax Output: Produces normalized class probabilities for each token, allowing straightforward decoding of POS tags.

2.1.1 Other models.

We tried other architectures such as using a simple LSTM layer, but the results were worse and as it can be seen in [1], we thought that a Bidirectional LSTM layer should fit better our task. We also tried a model with multiple Bidirectional LSTM Layers, but the results didn't improve, in fact they got worse and the training became more difficult as it will overfit earlier. The same result could be seen in [2].

3 Hyperparameter optimization

As we were training our models we tried different callbacks techniques such as Early stopping and Reduce on Plateau in order to obtain the best model in each training. We also try different embedding dimensions, LSTM units and batch sizes. If we use the callbacks, the number of epochs became irrelevant if we set it higher enough, as the training will stop automatically if no improvement is detected after a given number of epochs (patience).
We employed **sparse categorical cross-entropy** as the loss function, since POS labels are stored as integer indices. This choice avoids memory overhead from one-hot encoding and improves computational efficiency.
The batch size affects both performance and stability, if it's small it produces noisier gradients, but better generalization, and if it's larger we get smoother gradients, faster training, but potential overfitting.
Choosing an optimal learning rate is critical, as it balances speed and precision of convergence, especially for recurrent architectures like BiLSTM that are sensitive to gradient dynamics.
The embedding dimensionality determines how much semantic information each word representation can store. Lower values produce faster training, but risk of losing fine-grained semantic detail, and higher values give richer representations, but larger models and slower convergence.

The number of LSTM units defines the models representational capacity. Fewer units have less memory of temporal dependencies and lower computational cost, in contrast, more units give better ability to capture long-range dependencies but higher risk of overfitting and slower training. In our experiments, we tuned this parameter to balance performance and GPU memory limits. For the embedding dimension and LSTM units we got the best performance when they are greater than 100, so we set both to 128.

4 Results

The results of training our final models can be seen in the html files provided. Here we show the "Accuracy" value for each language:

	English	**Spanish**	**Italian**
Accuracy	0.9232	0.9381	0.9300

5 Conclusion

We developed a model that performs well for PoS tagging in the three languages studied, with an accuracy over 90%. However, what we can highlight most is the process that led us to develop the model. Implementing different architectures and using different techniques when training the models, as well as the different ways of working with the text to introduce it into our model. Our experiments demonstrated that the architecture could generalize well across English, Spanish, and Italian, highlighting its robustness even when trained under hardware constraints.
Beyond the empirical results, this work reinforced several key lessons from the lab sequence:

- The importance of bidirectional context for accurate sequence labeling.
- The role of embedding representations in handling lexical variability.
- The influence of training hyperparameters such as batch size, learning rate, and callback techniques such as early stopping on performance and convergence.

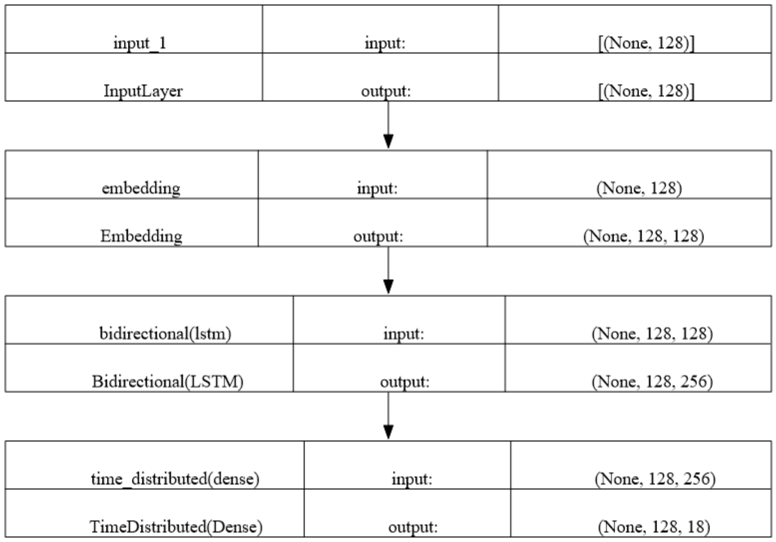

Figure 1: Schema of the final model architecture.

References

[1] Talukdar, K., Sarma, S. K., & Bhuyan, M. P. (2023). Parts of Speech (PoS) and Universal Parts of Speech (UPOS) Tagging: A Critical Review with Special Reference to Low Resource Languages.

[2] Wang, P., Qian, Y., Soong, F. K., He, L., & Zhao, H. (2015). A Unified Tagging Solution: Bidirectional LSTM Recurrent Neural Network with Word Embedding. *arXiv preprint arXiv:1511.00215*.